

Lecture 2: Enhanced Sampling and Free-Energy Calculations

1. Why Do We Need Enhanced Sampling?

In an equilibrium molecular simulation at temperature T , configurations $\mathbf{x}$ are distributed according to the Boltzmann distribution,

$$p(\mathbf{x}) = \frac{1}{Z} e^{-\beta U(\mathbf{x})}. \quad (1)$$

Molecular dynamics can, in principle, generate samples from this distribution. However, sampling may become inefficient when important regions of configuration space are separated by large free-energy barriers.

For example, suppose two metastable states A and B are separated by a free-energy barrier $\Delta F^\ddagger$. The probability of accessing configurations near the barrier is suppressed approximately as

$$P_{\text{barrier}} \sim e^{-\beta \Delta F^\ddagger}. \quad (2)$$

A large barrier therefore leads to long waiting times between transitions. Even if the simulation is formally correct, a finite trajectory may remain trapped in one metastable state.

This motivates a central idea in enhanced sampling:

sample from a modified distribution and reweight the samples afterward

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This same idea underlies free-energy perturbation, umbrella sampling, WHAM, BAR, and MBAR.

2. Free Energy and the Partition Function

The Helmholtz free energy is related to the canonical partition function by

$$F = -k_B T \ln Z. \quad (4)$$

For two thermodynamic states A and B ,

$$\Delta F_{BA} = F_B - F_A = -k_B T \ln \frac{Z_B}{Z_A}. \quad (5)$$

The absolute partition function is generally difficult to evaluate because it requires an integral over a high-dimensional configuration space. Free-energy calculations therefore usually focus on ratios of partition functions.

3. Free-Energy Perturbation

Suppose two systems have potential energies

$$U_A(\mathbf{x}) \quad (6)$$

and

$$U_B(\mathbf{x}). \quad (7)$$

Their partition functions are

$$Z_A = \int d\mathbf{x} e^{-\beta U_A(\mathbf{x})}, \quad (8)$$

$$Z_B = \int d\mathbf{x} e^{-\beta U_B(\mathbf{x})}. \quad (9)$$

Consider their ratio,

$$\frac{Z_B}{Z_A} = \frac{\int d\mathbf{x} e^{-\beta U_B(\mathbf{x})}}{\int d\mathbf{x} e^{-\beta U_A(\mathbf{x})}}. \quad (10)$$

Inside the numerator, multiply and divide by $e^{-\beta U_A(\mathbf{x})}$:

$$Z_B = \int d\mathbf{x} e^{-\beta U_A(\mathbf{x})} e^{-\beta [U_B(\mathbf{x}) - U_A(\mathbf{x})]}. \quad (11)$$

Therefore,

$$\frac{Z_B}{Z_A} = \left\langle e^{-\beta [U_B(\mathbf{x}) - U_A(\mathbf{x})]} \right\rangle_A. \quad (12)$$

Defining

$$\Delta U(\mathbf{x}) = U_B(\mathbf{x}) - U_A(\mathbf{x}), \quad (13)$$

we obtain the Zwanzig free-energy perturbation equation,

$$\boxed{\Delta F_{BA} = -k_B T \ln \left\langle e^{-\beta \Delta U} \right\rangle_A}. \quad (14)$$

This equation allows the free-energy difference between two states to be calculated using configurations sampled only from state A .

4. Free-Energy Perturbation as Reweighting

The FEP equation illustrates the more general idea of importance sampling.

Suppose configurations are sampled from a distribution

$$p_A(\mathbf{x}) = \frac{e^{-\beta U_A(\mathbf{x})}}{Z_A}. \quad (15)$$

The probability distribution of state B is

$$p_B(\mathbf{x}) = \frac{e^{-\beta U_B(\mathbf{x})}}{Z_B}. \quad (16)$$

Their probability ratio is

$$\frac{p_B(\mathbf{x})}{p_A(\mathbf{x})} = \frac{Z_A}{Z_B} e^{-\beta [U_B(\mathbf{x}) - U_A(\mathbf{x})]}. \quad (17)$$

Thus a configuration sampled from state A can be assigned a weight appropriate for state B :

$$w_{A \rightarrow B}(\mathbf{x}) \propto e^{-\beta \Delta U(\mathbf{x})}. \quad (18)$$

For an observable $O(\mathbf{x})$,

$$\langle O \rangle_B = \frac{\langle O(\mathbf{x})e^{-\beta\Delta U(\mathbf{x})} \rangle_A}{\langle e^{-\beta\Delta U(\mathbf{x})} \rangle_A}. \quad (19)$$

The same configurations can therefore represent different ensembles by assigning them different statistical weights.

5. The Overlap Problem

Although free-energy perturbation is formally exact, it can be statistically inefficient.

Suppose state A samples configurations from a region of phase space where state B has very little probability. The configurations important for Z_B may then be extremely rare in an A -state simulation.

Schematically,

$$p_A(\mathbf{x}) \not\approx p_B(\mathbf{x}). \quad (20)$$

The exponential average

$$\langle e^{-\beta\Delta U} \rangle_A \quad (21)$$

can then be dominated by a very small number of rare configurations.

The practical requirement for accurate reweighting is therefore

$$\boxed{\text{good overlap between the sampled and target distributions.}} \quad (22)$$

This observation motivates the introduction of intermediate states and biased simulations.

6. Umbrella Sampling

A common enhanced-sampling approach introduces a bias along a collective variable or reaction coordinate

$$\xi = \xi(\mathbf{x}). \quad (23)$$

For example, a harmonic umbrella centered at ξ_i may be written as

$$V_i(\xi) = \frac{k_i}{2}(\xi - \xi_i)^2. \quad (24)$$

The potential sampled in umbrella window i is then

$$U_i(\mathbf{x}) = U_0(\mathbf{x}) + V_i(\xi(\mathbf{x})). \quad (25)$$

Different windows are centered at different values of ξ ,

$$\xi_1, \xi_2, \dots, \xi_M, \quad (26)$$

so that the combined simulations cover a broad range of the reaction coordinate.

Each individual window samples

$$p_i(\mathbf{x}) = \frac{1}{Z_i} e^{-\beta U_i(\mathbf{x})}. \quad (27)$$

The central statistical problem is now:

How should samples from many different biased distributions be combined to estimate the unbiased ensemble?

7. Reduced Potentials

It is convenient to define the dimensionless reduced potential of state i :

$$u_i(\mathbf{x}) = \beta U_i(\mathbf{x}). \quad (28)$$

The corresponding partition function is

$$Z_i = \int d\mathbf{x} e^{-u_i(\mathbf{x})}. \quad (29)$$

Define the dimensionless free energy

$$f_i = -\ln Z_i. \quad (30)$$

Then the normalized distribution of state i is

$$p_i(\mathbf{x}) = e^{f_i - u_i(\mathbf{x})}. \quad (31)$$

Only differences between the f_i 's are physically meaningful because an overall additive free-energy constant is arbitrary.

8. Pooling Samples from Many States

Suppose we have M thermodynamic states and collect

$$N_i \quad (32)$$

samples from state i .

The total number of configurations is

$$N = \sum_{i=1}^M N_i. \quad (33)$$

We pool all samples,

$$\{\mathbf{x}_1, \dots, \mathbf{x}_N\}. \quad (34)$$

Every sampled configuration can, in principle, be evaluated using the potential-energy function of every thermodynamic state:

$$u_1(\mathbf{x}_n), u_2(\mathbf{x}_n), \dots, u_M(\mathbf{x}_n). \quad (35)$$

This is a key idea in MBAR:

every configuration contains information about every state.

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9. The Mixture Distribution

The pooled data may be regarded as being sampled from a mixture of the individual state distributions.

The probability density of the pooled ensemble is

$$p_{\text{mix}}(\mathbf{x}) = \sum_{k=1}^M p(\mathbf{x}, \lambda = k) = \sum_{k=1}^M p(\mathbf{x}|\lambda = k)p(\lambda = k) = \sum_{k=1}^M \frac{N_k}{N} p_k(\mathbf{x}). \quad (37)$$

Using

$$p_k(\mathbf{x}) = e^{f_k - u_k(\mathbf{x})}, \quad (38)$$

we obtain

$$p_{\text{mix}}(\mathbf{x}) = \frac{1}{N} \sum_{k=1}^M N_k e^{f_k - u_k(\mathbf{x})}. \quad (39)$$

For a configuration $\mathbf{x}$, the relative importance of state i compared with the pooled ensemble is therefore

$$\frac{p_i(\mathbf{x})}{p_{\text{mix}}(\mathbf{x})} = \frac{N e^{f_i - u_i(\mathbf{x})}}{\sum_{k=1}^M N_k e^{f_k - u_k(\mathbf{x})}}. \quad (40)$$

This ratio forms the basis of the MBAR weights.

10. MBAR Self-Consistency Equations

The partition function of state i can be written using importance sampling from the pooled ensemble:

$$Z_i = \int d\mathbf{x} p_{\text{mix}}(\mathbf{x}) \frac{e^{-u_i(\mathbf{x})}}{p_{\text{mix}}(\mathbf{x})}. \quad (41)$$

Approximating this integral using the N pooled samples gives

$$Z_i \approx \frac{1}{N} \sum_{n=1}^N \frac{e^{-u_i(\mathbf{x}_n)}}{p_{\text{mix}}(\mathbf{x}_n)}. \quad (42)$$

Substituting the expression for the mixture distribution,

$$p_{\text{mix}}(\mathbf{x}_n) = \frac{1}{N} \sum_{k=1}^M N_k e^{f_k - u_k(\mathbf{x}_n)}, \quad (43)$$

we obtain

$$e^{-f_i} = \sum_{n=1}^N \frac{e^{-u_i(\mathbf{x}_n)}}{\sum_{k=1}^M N_k e^{f_k - u_k(\mathbf{x}_n)}}. \quad (44)$$

Taking the negative logarithm yields the MBAR self-consistency equation,

$$\boxed{f_i = -\ln \sum_{n=1}^N \frac{e^{-u_i(\mathbf{x}_n)}}{\sum_{k=1}^M N_k e^{f_k - u_k(\mathbf{x}_n)}}}. \quad (45)$$

The equations for all f_i 's are coupled and are therefore solved simultaneously.